David Mingfei Liu

Contact Information

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Academic Appointments

Cornell University (2025 – Present)
Assistant Research Professor[1]
Center for Data Science for Enterprise and Society
Mentors: Jon Kleinberg, Maryam Fazel, and Sarah Dean
Faculty Affiliate, Cornell Center on Democracy

Education

Northeastern University (2025)
Ph.D., Computer Science
Khoury College of Computer Sciences and Network Science Institute
Advisor: Tina Eliassi-Rad

Princeton University (2018)
B.S.E. in Computer Science with Certificate in Statistics and Machine Learning
summa cum laude

Conference Papers

David Liu, Erik Weis, Moritz Laber, Tina Eliassi-Rad, and Brennan Klein. 2026. Identifying and Upweighting Power-Niche Users to Mitigate Popularity Bias in Recommendations. In *Proceedings of the ACM Web Conference 2026 (WWW)*. [pdf]

David Liu, Arjun Seshadri, Tina Eliassi-Rad, and Johan Ugander. 2025. Bypassing Skip-Gram Negative Sampling: Dimension Regularization as a More Efficient Alternative for Graph Embeddings. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD)*. [pdf]

David Liu, Jackie Baek, and Tina Eliassi-Rad. 2025. When Collaborative Filtering is not Collaborative: Unfairness of PCA for Recommendations. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. [pdf]

David Liu, Virginie Do, Nicolas Usunier, and Maximilian Nickel. 2023. Group fairness without demographics using social networks. In *2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*. [pdf]

David Liu and Tina Eliassi-Rad. 2023. STABLE: Identifying and Mitigating Instability in Embeddings of the Degenerate Core. In *Proceedings of the 2023 SIAM International Conference on Data Mining (SDM)*. [pdf]

Agapi Rissaki[*], Bruno Scarone*, **David Liu**, Aditeya Pandey, Brennan Klein, Tina Eliassi-Rad, and Michelle A. Borkin. 2022. BiaScope: Visual Unfairness Diagnosis for Graph Embeddings. In *2022 IEEE Visualization in Data Science (VDS)*. [pdf]

[1] A three-year non-tenure-track faculty position
[*] Equal contribution.

David Liu*, Priyanka Nanayakkara*, Sarah Ariyan Sakha, Grace Abuhamad, Su Lin Blodgett, Nicholas Diakopoulos, Jessica R. Hullman, and Tina Eliassi-Rad. 2022. Examining Responsibility and Deliberation in AI Impact Statements and Ethics Reviews. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*. [pdf]

David Liu*, Zohair Shafi*, William Fleisher, Tina Eliassi-Rad, and Scott Alfeld. 2021. RAWLSNET: Altering Bayesian Networks to Encode Rawlsian Fair Equality of Opportunity. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*. [pdf]

Journal Papers

Samantha Dies, **David M. Liu**, and Tina Eliassi-Rad. 2026. Forecasting faculty placement from patterns in coauthorship networks. *EPJ Data Science* 15, Article 41 (2026). [pdf]

David Liu and Matthew J. Salganik. 2019. Successes and struggles with computational reproducibility: Lessons from the Fragile Families Challenge. *Socius* 5 (2019). [pdf]

Workshop Papers

David Liu, Haruka Kiyohara, and Sarah Dean. 2026. Data Mixing for Group Preference Heterogeneity in Collaborative Filtering. In *Pluralistic Alignment Workshop at the International Conference on Machine Learning (ICML)*.

David Liu, Kiran Tomlinson, Jon Kleinberg, and Johan Ugander. 2026. Forecasting Instant Runoff Voting from Polls. In *3rd Workshop on New Directions in Social Choice at the ACM Conference on Economics and Computation (EC)*.

Public Scholarship

Cynthia Conti-Cook, Roya Pakzad, Sarah Ariyan Sakha, and **David Liu**. 2023. A Guiding Framework for Vetting Technology Vendors Operating in the Public Sector. A research report by the Ford Foundation. [link]

David M. Liu and Sarah Sakha. 2021. A New AI Lexicon: Power. The exclusionary costs of large-scale computing. AI Now Institute. [link]

Tutorials

2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT): Translation/Dialogue Tutorial: A Guiding Framework for Vetting Technology Vendors Operating in the Public Sector (with Cynthia Conti-Cook, Roya Pakzad, and Sarah Ariyan Sakha), Chicago, IL, June 2023.

Posters

David Liu. Disaggregating bike-share mobility data to measure urban segregation over time. *The 9th International Conference on Computational Social Science* (IC2S2), Copenhagen, Denmark, July 2023.

David Liu and Tina Eliassi-Rad. Increasing the Stability of Embeddings of the Degenerate Core. *The 2022 International Conference on Network Science* (NetSci), Virtual Conference, July 2022. (oral presentation)

David Liu. Uncovering the Evolution of Mukbang Videos on YouTube. *The 7th International Conference on Computational Social Science* (IC2S2), Virtual Conference, July 2021.

David Liu. Uncovering the Evolution of Mukbang Videos on YouTube. *Networks 2021: A Joint Sunbelt and NetSci Conference*, Virtual Conference, July 2021. (poster presentation)

Professional Experience

Meta Central Applied Science (CAS) May 2024 – Aug 2024
Research Scientist Intern
Graph Science and Statistics Research

University of California, Berkeley Fall 2022
Visiting Graduate Student
Simons Institute Program on "Graph Limits and Processes on Networks: From Epidemics to Misinformation"

Meta AI (FAIR) May 2022 – Aug 2022
AI Research Intern
Supervisor: Maximilian Nickel

Taraaz Feb 2022 – Aug 2022, Oct 2023 – Dec 2023
Researcher

Bloomberg LP Jul 2018 – Sep 2020
Software Engineer

The New York Times Jun 2017 – Aug 2017
Software Intern

Teaching

DS 4400: Machine Learning and Data Mining I Spring 2024
Northeastern University – *Instructor of Record*
4 credits, 86 undergraduate students, syllabus

Awards and Honors

2023 SDM Doctoral Forum and Travel Award
2021 NSF Graduate Research Fellowship (GRFP)
2018 Sigma Xi Nomination

Invited Talks

2026 CU Boulder Information Science Seminar. Boulder, CO.
2024 Brown University CNTR Tech and Society Seminar. Providence, RI.
2024 Tsinghua University. Beijing, China.
2024 MSR New England ML Ideas Seminar. Cambridge, MA.
2021 MD4SG Working Group on Discrimination in Algorithmic Decision-making. Online.
2019 RabbitMQ Summit, Keynote Speaker. *Growing a farm of rabbits to scale financial applications*. London, UK.
2017 Fragile Families Challenge Scientific Workshop. Princeton, NJ.